E-Commerce Sales & Customer Behaviour Analysis

Business Insights

1. Regional Performance: The North region recorded the highest sales among the regions analysed, making it an important market to review for continued growth opportunities.

2. Product Category Performance: Electronics generated the highest sales and total profit among product categories, indicating strong performance in both revenue generation and overall profit contribution.

3. Sales and Profit Relationship: Sales and Profit showed a strong positive correlation of 0.89. In this dataset, higher sales values generally tended to be associated with higher profit values.

4. Discount and Profitability: Average profit declined from approximately 11,564 at a 0% discount to approximately 7,184 at a 50% discount. This pattern suggests that larger discounts may reduce average profitability.

5. Customer Segment — Average Order Value: The Corporate customer segment recorded the highest average order value, at approximately 52,158, indicating a higher average transaction value than the other segments.

6. Shipping Cost and Profit: Shipping_Cost and Profit showed almost no linear correlation in the analysis. Shipping cost alone may therefore not explain profit differences; review it alongside sales, quantity, product and order details.

7. Sales Distribution: Sales had positive (right) skewness, with a skewness value of approximately 2.12. This indicates that most transactions were concentrated at lower sales values, while a smaller number of high-value transactions created a long right tail.

8. Profit Distribution and Data Quality: Profit values contained extreme observations, including records with Profit equal to 500,000. These records were treated as potential data-quality issues during cleaning and should be verified against original transactions.